Year 12 Maths - Trigonometric functions and equations

Similar questions - Trigonometric graphs and models

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 The depth of water, d metres, in a harbour is modelled by $d = 7 - 3\cos(30t)^\circ$, where t is the number of hours after midnight and $0 \leqslant t \leqslant 24$.

(a) Write down the least depth and the greatest depth predicted by the model, and the times of the first low tide and first high tide. **[3]**

(b) Find the time interval between consecutive high tides. **[1]**

(c) A ship needs a depth of at least 8.5 m to enter the harbour. Find the times, in hours after midnight, between which the ship can enter on the first of the two occasions. **[4]**

(d) State one way in which the model may be unrealistic. **[1]**

2 A curve has equation $y = 1 + 3\sin(2x + 40)^\circ$ for $0 \leqslant x \leqslant 360$.

(a) Write down the maximum and minimum values of y. **[2]**

(b) Find the smallest positive value of x at which the maximum occurs. **[2]**

(c) Solve $y = 2.5$, giving all solutions in the given range. **[4]**

(d) How many times does the curve cross the line $y = 1$ in this range? Give a reason. **[2]**

Question 1

(a) Least depth 4 m, greatest depth 10 m. First low tide at $t = 0$ (midnight), first high tide when $30t = 180$, i.e. $t = 6$ (6 am)

(b) The period is $\dfrac{360}{30} = 12$ hours

(c) $7 - 3\cos(30t) \geqslant 8.5$ gives $\cos(30t) \leqslant -\frac{1}{2}$, so $120 \leqslant 30t \leqslant 240$ and $4 \leqslant t \leqslant 8$. The ship can enter between 4 am and 8 am

(d) E.g. the real tide times shift from day to day (the period is about 12.4 hours), and real tides depend on weather and the Moon, so the same pattern would not repeat exactly

Question 2

(a) Maximum 4, minimum -2

(b) $2x + 40 = 90$ gives $x = 25$

(c) $\sin(2x + 40) = 0.5$. With $40 \leqslant 2x + 40 \leqslant 760$: $2x + 40 = 150,\ 390,\ 510,\ 750$ (and 30 is outside the range). So $x = 55,\ 175,\ 235,\ 355$

(d) $y = 1$ when $\sin(2x + 40) = 0$, i.e. $2x + 40 = 180, 360, 540, 720$ giving $x = 70, 160, 250, 340$. Four crossings